

PathFinder

Chatbot Intent Catalogue

Excluding SAE / Dashboard Analytics

Overview

PathFinder currently supports 26 locked chatbot intents grouped into 7 domains. Each intent is mapped to a specific academic or career advising function and is handled by a dedicated engine: the Knowledge Graph (KG) Engine, the Retrieval-Augmented Generation (RAG) Engine, or the Academic Logic Engine (ALE). The intents below represent the full, locked scope of the PathFinder chatbot as of the current system version.

#	Domain	Intents	Intent Names
1	Student Record	1	<i>get_student_record</i>
2	Policy / Handbook Questions	1	<i>policy_query</i>
3	Course Information	4	<i>get_course_info, get_course_prerequisites, get_skills_taught, search_courses_by_skill</i>
4	Career / Role Guidance	8	<i>get_role_profile, get_roles_by_track, compute_skill_gap, compute_alignment_score, recommend_courses_to_close_gap, find_best_matching_roles, estimate_alignment_improvement, get_focus_courses_for_target</i>
5	Track Guidance	4	<i>get_track_overview, compare_tracks, recommend_track_for_role, recommend_track_for_skill</i>
6	Academic Planning	6	<i>plan_semester, generate_graduation_roadmap, run_graduation_audit, check_course_eligibility, simulate_gpa_forward, solve_target_gpa</i>
7	Control / Scope Handling	2	<i>clarification_needed, out_of_scope</i>
	Total	26	26 locked chatbot intents across 7 domains

1. Student Record

This domain answers questions about the logged-in student's academic record using the Student Context Provider. It covers personal academic data such as GPA, academic level, standing, current and completed courses, failed courses, track enrollment, and academic warnings. It also handles what-if assumption acknowledgement and resetting assumptions.

Intent	What It Does	Example Student Question
<code>get_student_record</code>	Retrieves student-specific academic record information such as CGPA, academic level, academic standing, current courses, completed courses, failed courses, track, warnings, and course status. It also supports what-if assumption acknowledgement and resetting assumptions.	"What is my current level?"

2. Policy / Handbook Questions

This domain answers university regulation and handbook questions using the RAG (Retrieval-Augmented Generation) engine with citations drawn directly from the official CIS Student Handbook. It covers attendance, grading, warnings, retake rules, graduation and honors requirements, and credit registration limits.

Intent	What It Does	Example Student Question
<code>policy_query</code>	Answers policy, regulation, and handbook questions such as attendance, grading scale, academic warnings, retakes, honors, graduation requirements, credit limits, and related academic rules.	"What is the attendance policy?"

3. Course Information

This domain answers curriculum and course catalogue questions using structured course data from the Knowledge Graph. It covers course details, prerequisite chains, skills taught per course, and cross-course skill searches.

Intent	What It Does	Example Student Question
<code>get_course_info</code>	Provides details about a course such as name, code, credits, level, offering semester, description, and related track information when available.	"Tell me about Advanced Programming."
<code>get_course_prerequisites</code>	Returns the prerequisites of a course. It can answer direct prerequisites or the full prerequisite chain depending on the question.	"What are the full prerequisites of NLP?"
<code>get_skills_taught</code>	Shows the skills taught by a specific course.	"What skills does Computer Vision teach?"
<code>search_courses_by_skill</code>	Finds courses that teach a specific skill or topic.	"Which courses teach machine learning?"

4. Career / Role Guidance

This domain connects the curriculum to career roles using course–skill–role mappings in the Knowledge Graph. Student-aware intents use the student's completed courses to estimate skill coverage and career role alignment. It helps students identify skill gaps, evaluate career fit, and plan course selections to improve role readiness.

Intent	What It Does	Example Student Question
get_role_profile	Provides information about a career role, including required skills and role-related details.	"Tell me about the Data Scientist role."
get_roles_by_track	Lists career roles connected to a specific academic track based on the track's courses and taught skills.	"What roles are connected to the AI track?"
compute_skill_gap	Compares the student's completed-course skills against the skills required for a target role and identifies missing skills.	"I want to become a Data Scientist. What am I missing?"
compute_alignment_score	Calculates how aligned the student's current completed courses are with a specific career role.	"How aligned am I with the Data Scientist role?"
recommend_courses_to_close_gap	Recommends courses that can help the student close the skill gap for a target role.	"What courses should I take to close my gap for Data Scientist?"
find_best_matching_roles	Ranks career roles based on how well they match the student's completed courses and acquired skills.	"What roles fit me best based on my completed courses?"
estimate_alignment_improvement	Estimates how much the student's alignment with a target role would improve if they took or passed specific planned courses.	"If I take Machine Learning and Data Mining, how much better would my Data Scientist alignment be?"
get_focus_courses_for_target	Identifies important or focus courses related to a target role or track. It is career and curriculum guidance, not an official semester plan.	"What future courses should I focus on for ML Engineer?"

5. Track Guidance

This domain helps students understand and compare the academic tracks offered by the CIS program. It maps tracks to associated skills and career roles, enabling students to evaluate which track best fits their academic interests or career goals.

Intent	What It Does	Example Student Question
<code>get_track_overview</code>	Provides an overview of an academic track, including its courses, skills, and related curriculum structure.	"Give me an overview of the AI track."
<code>compare_tracks</code>	Compares two academic tracks side by side.	"Compare AI and Data Science."
<code>recommend_track_for_role</code>	Recommends or ranks tracks based on their relevance to a target career role.	"Which track is best for Data Scientist?"
<code>recommend_track_for_skill</code>	Recommends or ranks tracks based on their relevance to a target skill.	"Which track is best for learning machine learning?"

6. Academic Planning

This domain covers student-specific academic decision-making and planning. These intents depend on the student's academic record and integrate with the Academic Logic Engine (ALE) where rule-driven computation is required. It supports semester planning, multi-semester roadmaps, graduation audits, course eligibility checks, and GPA simulation.

Intent	What It Does	Example Student Question
plan_semester	Generates a suggested course plan for the next semester only, based on the student's academic record, completed courses, failed courses, track, CGPA, credit limits, and course availability.	"What courses should I take next semester?"
generate_graduation_roadmap	Builds a multi-semester roadmap showing how the student can progress from the current academic state until graduation, or until a specific target semester if the student asks for one.	"Can I graduate by Fall 2027?"
run_graduation_audit	Checks whether the student currently satisfies graduation requirements, identifies missing graduation requirements, and includes honors eligibility and audit information where applicable.	"Can I graduate?"
check_course_eligibility	Checks whether the student can take a specific course, mainly based on completed prerequisites, current enrollment, previous completion, failed/retake status, and relevant course registration conditions.	"Can I take Computer Vision next semester?"
simulate_gpa_forward	Simulates the student's future CGPA based on hypothetical grades in current or planned courses.	"If I get A in Machine Learning and C in Big Data Engineering, what will my CGPA be?"
solve_target_gpa	Calculates what grades the student needs in upcoming or current courses to reach a target CGPA.	"What grades do I need to reach a 2.0 CGPA?"

7. Control / Scope Handling

These intents handle edge cases where the system should not directly invoke an academic engine. They ensure graceful handling of ambiguous queries, missing required entities, and out-of-scope requests that fall outside PathFinder's advising domain.

Intent	What It Does	Example Student Question
clarification_needed	Used when the question is ambiguous, missing a required entity, or could refer to multiple valid academic or career meanings.	"Tell me about software engineering."
out_of_scope	Used when the question is outside PathFinder's academic advising, curriculum, policy, career, or student-progress scope.	"Where is the cafeteria?"

Note: All 26 intents listed above are locked. No modifications to the intent taxonomy, routing logic, or engine assignments are permitted without a formal design review and update to the locked design documents.